b. Show that the vector field $\vec{F}=\left(x^2-y^2+x\right)\vec{i}-\left(2xy+y\right)\vec{j}$ is irrotational and hence find its scalar potential. | 8 | 4 | 2 | 2

23. a. Find the laplace transform of the function $f(t)=\begin{cases} t \text{ for } 0<t<a \\ 2a-t \text{ for } a<t<2a \end{cases}$ where $f(t+2a)=f(t)$. | 8 | 4 | 3 | 2

(OR)

b. Solve $\frac{d^2y}{dx^2}+6\frac{dy}{dx}+8y=2$ given $y(0)=1, y'(0)=1$. | 8 | 4 | 3 | 2

24. a. If $f(z)=u+iv$ is analytic find f(z) and v if $u=\frac{\sin 2x}{\cos 2x+\cosh 2y}$. | 8 | 3 | 4 | 1

(OR)

b. Find the bilinear transformation which maps the points z = –2, 0, 2 into the points ω=0, i, –i respectively. | 8 | 3 | 4 | 2

25. a.i. Using Cauchy integral formula evaluate $\int_c \frac{z}{z-2}dz$, where c is the circle $|z-2|=3/2$. | 4 | 3 | 5 | 1

ii. Evaluate $\int_c \frac{\sin \pi z^2+\cos \pi z^2}{(z-1)(z-2)}dz$ where C:|z|=3 by Cauchy's residue theorem. | 4 | 3 | 5 | 1

(OR)

b. Expand $f(z)=\frac{z^2-1}{(z+2)(z+3)}$ in a Laurent's series if | 8 | 4 | 5 | 2

(i) $|z|<2$
(ii) $|z|>3$
(iii) $2<|z|<3$

PART – C (1 × 15 = 15 Marks)

Answer **ANY ONE** Question

	Marks	BL	CO	PO

26. Verify Gauss divergence theorem for $\vec{F}=\left(x^2-yz\right)\vec{i}+\left(y^2-zx\right)\vec{j}+\left(z^2-xy\right)\vec{k}$ taken over the rectangular parallelopiped enclosed by $x=0, x=a, y=0, y=b, z=0$ and $z=c$. | 15 | 4 | 2 | 2

27. Evaluate $\int_0^{2\pi} \frac{d\theta}{13+5\sin\theta}$ using contour integration. | 15 | 4 | 5 | 2

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Reg. No.															

B.Tech. / M.Tech (Integrated) DEGREE EXAMINATION, MAY 2023
Second Semester

21MAB102T – ADVANCED CALCULUS AND COMPLEX ANALYSIS
(For the candidates admitted from the academic year 2021 - 2022 & 2022 - 2023)

Note:
(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part – B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours | Max. Marks: 75

PART – A (20 × 1 = 20Marks)

Answer **ALL** Questions

	Marks	BL	CO	PO

1. Evaluation of $\int_0^2\int_0^2 dxdy$ is | 1 | 2 | 1 | 1

(A) 0 (B) 1
(C) 4 (D) 2

2. Area of the double integral in Cartesian co-ordinate is | 1 | 1 | 1 | 2

(A) $\iint dydx$ (B) $\iint xdxdy$
(C) $\iint x^2dxdy$ (D) $\iint rdrd\theta$

3. The region of integration of the integral $\int_0^1\int_0^x f(x,y)dxdy$ is | 1 | 2 | 1 | 1

(A) Square (B) Rectangle
(C) Triangle (D) Circle

4. Evaluation of $\int_0^1\int_0^2\int_0^3 dxdydz$ is equal to | 1 | 2 | 1 | 1

(A) 3 (B) 4
(C) 2 (D) 6

5. If $\vec{r}$ is the position vector of the point (x,y,z) with respect to the origin then $\nabla.\vec{r}$ is | 1 | 1 | 2 | 1

(A) 2 (B) 3
(C) 0 (D) 1

6. If $\phi=xyz$ then $\nabla\phi$ is | 1 | 2 | 2 | 2

(A) $yz\vec{i}+zx\vec{j}+xy\vec{k}$ (B) $xy\vec{i}+yz\vec{j}+zx\vec{k}$
(C) $zx\vec{i}+xy\vec{j}+yz\vec{k}$ (D) 0

THE HELPERS

7.	The unit normal vector to the surface $x^2+y^2-z^2=1$ *at* $(1,1,1)$ is			1	2	2	1
	(A) $(\vec{i}+\vec{j}-\vec{k})/\sqrt{3}$	(B) $(2\vec{i}+2\vec{j}-2\vec{k})/\sqrt{2}$					
	(C) $(3\vec{i}+3\vec{j}-3\vec{k})/2\sqrt{3}$	(D) $(\vec{i}+\vec{j}-\vec{k})/3\sqrt{2}$					
8.	If $\vec{u}$ *and* $\vec{v}$ are irrotational then $\vec{u}\times\vec{v}$ is			1	1	2	1
	(A) Solenoidal	(B) Irrotational					
	(C) Constant vector	(D) Zero vector					
9.	Find $L[e^{-at}]$			1	1	3	1
	(A) $\frac{1}{s+1}$	(B) $\frac{1}{s-1}$					
	(C) $\frac{1}{s+a}$	(D) $\frac{1}{s-a}$					
10.	If $L[f(t)]=F(s)$ then $L[tf(t)]=$			1	1	3	2
	(A) $\frac{d}{ds}F(s)$	(B) $-\frac{d}{ds}F(s)$					
	(C) $(-1)^n\frac{d}{ds}F(s)$	(D) $-\frac{d^2}{ds^2}F(s)$					
11.	Find $L^{-1}\left[\frac{1}{(s-1)^2}\right]$			1	1	3	2
	(A) te^{-t}	(B) te^{t}					
	(C) t^2e^t	(D) t					
12.	Using the final value theorem, find the value of the function $f(t)=1+e^{-t}(\sin t+\cos t)$			1	2	3	2
	(A) 1	(B) 0					
	(C) ∞	(D) –2					
13.	If a function $u(x,y)$ satisfies $u_{xx}+u_{yy}=0$ then u is			1	1	4	1
	(A) Analytic	(B) Harmonic					
	(C) Differentiable	(D) Continuous					
14.	The points at which the function $f(z)=\frac{1}{z^2+1}$ fails to be analytic is			1	1	4	1
	(A) $z=\pm1$	(B) $z=0$					
	(C) $z=\pm i$	(D) $z=\pm2$					
15.	An analytic function with constant modulus is			1	1	4	1
	(A) Zero	(B) Analytic					
	(C) Constant	(D) Harmonic					

16.	The transformation ω=z+c where c is a complex constant represents			1	1	4	1
	(A) Rotation	(B) Magnification					
	(C) Translation	(D) Magnification and rotation					
17.	The value of $\int_c \frac{zdz}{z-2}$, where c is the circle |z|=1 is			1	2	5	2
	(A) 0	(B) $\frac{\pi}{2}i$					
	(C) π/2	(D) 2					
18.	Let $C_1:\|z-a\|=R_1$ *and* $C_2:\|z-a\|=R_2$ be two concentric circles ($R_2<R_1$), the annular region is defined as			1	1	5	1
	(A) Within C_1	(B) Within C_2					
	(C) Within C_2 and outside C_1	(D) Within C_1 and outside C_2					
19.	If $f(z)=\frac{\sin z}{z}$, then			1	1	5	1
	(A) z=0 is a simple pole	(B) z=0 is a pole of order 2					
	(C) z=0 is a removable singularity	(D) z=0 is a zero of f(z)					
20.	If $f(z)$ is analytic inside and on C, the value of $\int_c \frac{f(z)}{z-a}dz$ where C is the simple closed curve and 'a' is any point within C is			1	1	5	2
	(A) $f(a)$	(B) $2\pi i f(a)$					
	(C) $2\pi i f'(a)$	(D) 0					

PART – B (5 × 8 = 40 Marks)

Answer **ALL** Questions

		Marks	BL	CO	PO
21. a.	By changing the order of integration, evaluate $\int_0^{4a}\int_{x^2/4a}^{2\sqrt{ax}} dydx$.	8	4	1	2
	(OR)				
b.	Evaluate $\int_0^a\int_0^{\sqrt{a^2-x^2}}\int_0^{\sqrt{a^2-x^2-y^2}} \frac{dzdydx}{\sqrt{a^2-x^2-y^2-z^2}}$.	8	4	1	2
22. a.i.	Find the directional derivative of $\phi=x^2yz+4xz^2+xyz$ *at* $(1,2,3)$ in the direction of $2\vec{i}+\vec{j}-\vec{k}$.	4	3	2	1
ii.	Find the angle between the surfaces $z=x^2+y^2-3$ *and* $x^2+y^2+z^2=9$ *at* $(2,-1,2)$.	4	3	2	1
	(OR)				

THE HELP PAPERS